

analysis

2025-11-10

```
df_image <- read.csv("../..plot/processed_data_image.csv")
df_image$experiment_type <- "image"

df_text <- read.csv("../..plot/processed_data_text.csv")
df_text$experiment_type <- "text"

df <- bind_rows(df_image, df_text) %>%
  mutate(outcome_grouped = case_when(
    outcome <= 40 ~ '10-40',
    outcome == 50 ~ '50',
    outcome >= 60 ~ '60-90'
  ),
  outcome_grouped = factor(outcome_grouped,
    levels = c('50',
               '10-40',
               '60-90')),
  event_type = factor(event_type,
    levels = c('gumball',
               'election'))))

contrasts(df$outcome_grouped)
```

```
##      10-40 60-90
## 50         0    0
## 10-40      1    0
## 60-90      0    1
```

```
contrasts(df$event_type)
```

```
##      election
## gumball      0
## election     1
```

Image analyses

Might

```
might_image <- glmer(prob ~ outcome_grouped*event_type +
  (1 + outcome_grouped*event_type | random_id),
  family = binomial,
  data = df %>% filter(utterance == 'might', experiment_type == 'image'),
  glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))
```

```
## Warning in eval(family$initialize, rho): non-integer #successes in a binomial
## glm!
```

```
## boundary (singular) fit: see help('isSingular')
summary(might_image)

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: prob ~ outcome_grouped * event_type + (1 + outcome_grouped *
##          event_type | random_id)
## Data: df %>% filter(utterance == "might", experiment_type == "image")
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
##
##          AIC          BIC      logLik -2*log(L)  df.resid
##      5363.0      5544.7    -2654.5    5309.0      6165
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -26.0216  -0.3122   0.1014   0.5645   7.4213
##
## Random effects:
##   Groups      Name                Variance Std.Dev. Corr
##   random_id (Intercept)            3.731    1.932
##             outcome_grouped10-40    2.836    1.684    0.57
##             outcome_grouped60-90    1.964    1.401   -0.96
##             event_typeelection      3.288    1.813    0.07
##             outcome_grouped10-40:event_typeelection 6.486    2.547   -0.85
##             outcome_grouped60-90:event_typeelection 4.750    2.179   -0.32
##
##
##
## -0.54
## -0.45 -0.30
## -0.11 0.89 -0.53
## 0.31 0.52 -0.97 0.73
## Number of obs: 6192, groups: random_id, 86
##
## Fixed effects:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.2251    0.2738   0.822   0.4110
## outcome_grouped10-40      0.2823    0.2836   0.996   0.3194
## outcome_grouped60-90     -1.4153    0.2436  -5.810 6.25e-09
## event_typeelection      0.9815    0.3536   2.776   0.0055
## outcome_grouped10-40:event_typeelection -4.1130    0.4494  -9.152 < 2e-16
## outcome_grouped60-90:event_typeelection -1.8347    0.3965  -4.627 3.70e-06
##
## (Intercept)
## outcome_grouped10-40
## outcome_grouped60-90      ***
## event_typeelection      **
## outcome_grouped10-40:event_typeelection ***
## outcome_grouped60-90:event_typeelection ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##          (Intr) ot_10-40 ot_60-90 evnt_t o_10-40:
## otc_g10-40 -0.139
## otc_g60-90 -0.927  0.247
## evnt_typlct -0.272  0.140    0.239
## otc_10-40:_ -0.154 -0.409    0.070   -0.738
## otc_60-90:_  0.129 -0.159   -0.166   -0.945  0.772
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

Not

```
not_image <- glmer(prob ~ outcome_grouped*event_type +
                    (1 + outcome_grouped*event_type | random_id),
                    family = binomial,
                    data = df %>% filter(utterance == 'bare not', experiment_type == 'image'),
                    glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))
```

```
## Warning in eval(family$initialize, rho): non-integer #successes in a binomial
## glm!
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(not_image)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: prob ~ outcome_grouped * event_type + (1 + outcome_grouped *
##          event_type | random_id)
## Data: df %>% filter(utterance == "bare not", experiment_type == "image")
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
##
##          AIC          BIC    logLik -2*log(L)  df.resid
##      3409.2      3590.9   -1677.6    3355.2     6165
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.2223  -0.1657  -0.0752   0.3978   9.4880
##
## Random effects:
##      Groups      Name                Variance Std.Dev. Corr
## random_id (Intercept)
##              outcome_grouped10-40    0.03371  0.1836  -0.80
##              outcome_grouped60-90    2.08550  1.4441  -0.59
##              event_typeelection      2.91915  1.7086  -0.97
##              outcome_grouped10-40:event_typeelection 2.18253  1.4773   0.46
##              outcome_grouped60-90:event_typeelection 1.35848  1.1655   0.28
##
##
##      -0.01
##      0.85  0.46
##     -0.87  0.41 -0.49
##      0.26 -0.82 -0.05 -0.40
## Number of obs: 6192, groups: random_id, 86
```

```
##
## Fixed effects:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -3.6718    0.5229  -7.023 2.18e-12
## outcome_grouped10-40      2.8965    0.4779   6.061 1.35e-09
## outcome_grouped60-90     -1.5820    0.6526  -2.424  0.0153
## event_typeelection      -0.9214    0.7712  -1.195  0.2322
## outcome_grouped10-40:event_typeelection  3.2716    0.7716   4.240 2.23e-05
## outcome_grouped60-90:event_typeelection  2.0077    0.9013   2.228  0.0259
##
## (Intercept)          ***
## outcome_grouped10-40      ***
## outcome_grouped60-90      *
## event_typeelection
## outcome_grouped10-40:event_typeelection ***
## outcome_grouped60-90:event_typeelection *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) ot_10-40 ot_60-90 evnt_t o_10-40:
## otcm_g10-40 -0.904
## otcm_g60-90 -0.770  0.769
## evnt_typlct -0.679  0.638   0.497
## otc_10-40:_  0.612 -0.650  -0.448  -0.960
## otc_60-90:_  0.558 -0.584  -0.707  -0.815  0.800
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

Probably

```
probably_image <- glmer(prob ~ outcome_grouped*event_type +
  (1 + outcome_grouped*event_type | random_id),
  family = binomial,
  data = df %>% filter(utterance == 'probably', experiment_type == 'image'),
  glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))
```

```
## Warning in eval(family$initialize, rho): non-integer #successes in a binomial
## glm!
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(probably_image)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: prob ~ outcome_grouped * event_type + (1 + outcome_grouped *
## event_type | random_id)
## Data: df %>% filter(utterance == "probably", experiment_type == "image")
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    3966.9    4148.6   -1956.4    3912.9     6165
##
```

```

## Scaled residuals:
##      Min       1Q   Median       3Q      Max
##   -4.99   -0.13    0.19    0.53 1757.91
##
## Random effects:
##      Groups      Name                Variance Std.Dev. Corr
##   random_id (Intercept)             1.8628   1.3648
##               outcome_grouped10-40    0.7421   0.8615  -0.73
##               outcome_grouped60-90    1.2103   1.1001  -0.40
##               event_typeelection      65.7856   8.1108  -0.77
##               outcome_grouped10-40:event_typeelection 56.8659   7.5410   0.57
##               outcome_grouped60-90:event_typeelection 72.9573   8.5415   0.74
##
##
##
##   -0.34
##   1.00 -0.28
##  -0.98  0.52 -0.97
##  -1.00  0.32 -1.00  0.98
## Number of obs: 6192, groups:  random_id, 86
##
## Fixed effects:
##                                Estimate Std. Error z value Pr(>|z|)
## (Intercept)                   -2.1146     0.2531  -8.356 < 2e-16
## outcome_grouped10-40          -1.6829     0.3034  -5.547 2.90e-08
## outcome_grouped60-90           2.1400     0.2471   8.659 < 2e-16
## event_typeelection            -8.5609     1.5791  -5.421 5.92e-08
## outcome_grouped10-40:event_typeelection  8.1773     1.5901   5.143 2.71e-07
## outcome_grouped60-90:event_typeelection  9.2486     1.6145   5.728 1.01e-08
##
## (Intercept)                  ***
## outcome_grouped10-40         ***
## outcome_grouped60-90         ***
## event_typeelection           ***
## outcome_grouped10-40:event_typeelection ***
## outcome_grouped60-90:event_typeelection ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)  ot_10-40  ot_60-90  evnt_t  o_10-40:
## otc_m_g10-40 -0.686
## otc_m_g60-90 -0.792  0.525
## evnt_typlct -0.328  0.158   0.007
## otc_10-40:_  0.251 -0.213   0.047  -0.971
## otc_60-90:_  0.325 -0.163  -0.002  -0.997  0.970
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

```

Text analyses

Might

```
might_text <- glmer(prob ~ outcome_grouped*event_type +  
  (1 + outcome_grouped*event_type | random_id),  
  family = binomial,  
  data = df %>% filter(utterance == 'might', experiment_type == 'text'),  
  glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))
```

```
## Warning in eval(family$initialize, rho): non-integer #successes in a binomial  
## glm!
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(might_text)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace  
## Approximation) [glmerMod]  
## Family: binomial ( logit )  
## Formula: prob ~ outcome_grouped * event_type + (1 + outcome_grouped *  
## event_type | random_id)  
## Data: df %>% filter(utterance == "might", experiment_type == "text")  
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))  
##  
##      AIC      BIC    logLik -2*log(L)  df.resid  
##  3381.6   3551.0   -1663.8    3327.6     3897  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -1.8510 -0.0175  0.4284  0.8384  4.6950  
##  
## Random effects:  
##      Groups      Name                Variance Std.Dev. Corr  
## random_id (Intercept)          2.95987  1.7204  
##      outcome_grouped10-40        0.04678  0.2163  -0.49  
##      outcome_grouped60-90        3.65784  1.9125  -0.99  
##      event_typeelection          0.01710  0.1308   0.42  
##      outcome_grouped10-40:event_typeelection 0.51639  0.7186  -0.67  
##      outcome_grouped60-90:event_typeelection 0.24963  0.4996  -0.95  
##  
##  
##      0.58  
##      -1.00 -0.52  
##      -0.32  0.58  0.39  
##      0.20  0.91 -0.13  0.86  
## Number of obs: 3924, groups: random_id, 109  
##  
## Fixed effects:  
##  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)      -0.15747    0.25047  -0.629   0.5295  
## outcome_grouped10-40      -0.53068    0.20979  -2.530   0.0114  
## outcome_grouped60-90     -1.88166    0.28334  -6.641 3.12e-11  
## event_typeelection      -0.20998    0.26405  -0.795   0.4265  
## outcome_grouped10-40:event_typeelection -0.78313    0.30964  -2.529   0.0114
```

```
## outcome_grouped60-90:event_typeelection 0.09413 0.31202 0.302 0.7629
##
## (Intercept)
## outcome_grouped10-40 *
## outcome_grouped60-90 ***
## event_typeelection
## outcome_grouped10-40:event_typeelection *
## outcome_grouped60-90:event_typeelection
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) ot_10-40 ot_60-90 evnt_t o_10-40:
## otc_m_g10-40 -0.696
## otc_m_g60-90 -0.921 0.624
## evnt_typlct -0.508 0.616 0.444
## otc_10-40:_ 0.338 -0.671 -0.300 -0.846
## otc_60-90:_ 0.349 -0.522 -0.435 -0.845 0.745
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

Not

```
not_text <- glmer(prob ~ outcome_grouped*event_type +
                  (1 + outcome_grouped*event_type | random_id),
                  family = binomial,
                  data = df %>% filter(utterance == 'bare not', experiment_type == 'text'),
                  glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))
```

```
## Warning in eval(family$initialize, rho): non-integer #successes in a binomial
## glm!
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(not_text)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: prob ~ outcome_grouped * event_type + (1 + outcome_grouped *
##          event_type | random_id)
## Data: df %>% filter(utterance == "bare not", experiment_type == "text")
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
##
##          AIC          BIC    logLik -2*log(L)  df.resid
##      2378.5      2547.9   -1162.3    2324.5      3897
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
##      -2        0         0         1 582030
##
## Random effects:
##      Groups      Name                                Variance Std.Dev. Corr
##      random_id (Intercept)                    40.19      6.340
##                   outcome_grouped10-40        26.00      5.099   -1.00
```

```
##          outcome_grouped60-90          74.66      8.641      -1.00
##          event_typeelection          33.42      5.781      -1.00
##          outcome_grouped10-40:event_typeelection 35.85      5.988       1.00
##          outcome_grouped60-90:event_typeelection 54.09      7.354       0.98
##
##
##          0.99
##          0.99 1.00
##          -1.00 -0.99 -0.99
##          -0.96 -0.99 -0.99 0.96
## Number of obs: 3924, groups: random_id, 109
##
## Fixed effects:
##
##          Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -11.128486    0.001620   -6868   <2e-16
## outcome_grouped10-40     10.640743    0.001620    6567   <2e-16
## outcome_grouped60-90      3.806004    0.001620    2349   <2e-16
## event_typeelection        6.944110    0.001620    4286   <2e-16
## outcome_grouped10-40:event_typeelection -6.197599    0.001620   -3825   <2e-16
## outcome_grouped60-90:event_typeelection -4.269721    0.001673   -2552   <2e-16
##
## (Intercept)          ***
## outcome_grouped10-40          ***
## outcome_grouped60-90          ***
## event_typeelection          ***
## outcome_grouped10-40:event_typeelection ***
## outcome_grouped60-90:event_typeelection ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) ot_10-40 ot_60-90 evnt_t o_10-40:
## otc_m_g10-40 0.000
## otc_m_g60-90 0.000 0.000
## evnt_typlct 0.000 0.000 0.000
## otc_10-40:_ 0.000 0.000 0.000 0.000
## otc_60-90:_ 0.000 0.000 0.000 0.000 0.000
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

Probably

```
probably_text <- glmer(prob ~ outcome_grouped*event_type +
                      (1 + outcome_grouped*event_type | random_id),
                      family = binomial,
                      data = df %>% filter(utterance == 'probably', experiment_type == 'text'),
                      glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))

## Warning in eval(family$initialize, rho): non-integer #successes in a binomial
## glm!

## boundary (singular) fit: see help('isSingular')
```

```
summary(probably_text)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: prob ~ outcome_grouped * event_type + (1 + outcome_grouped *
##          event_type | random_id)
## Data: df %>% filter(utterance == "probably", experiment_type == "text")
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 1e+05))
##
##          AIC          BIC      logLik -2*log(L)  df.resid
##      3058.6      3228.0    -1502.3    3004.6     3897
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.192  -0.118   0.177   0.532  105.386
##
## Random effects:
##      Groups      Name                Variance Std.Dev. Corr
## random_id (Intercept)            0.000     0.000
##                outcome_grouped10-40      1.022     1.011    NaN
##                outcome_grouped60-90      1.055     1.027    NaN
##                event_typeelection        2.551     1.597    NaN
##                outcome_grouped10-40:event_typeelection 2.724     1.650    NaN
##                outcome_grouped60-90:event_typeelection 2.666     1.633    NaN
##
##
##      -0.68
##      -0.17 -0.61
##       0.19  0.59 -1.00
##       0.01  0.73 -0.99  0.98
## Number of obs: 3924, groups:  random_id, 109
##
## Fixed effects:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -2.6791     0.2762  -9.698  <2e-16
## outcome_grouped10-40      -0.5349     0.3587  -1.491  0.1359
## outcome_grouped60-90       2.7113     0.3031   8.946  <2e-16
## event_typeelection      -1.0404     0.5148  -2.021  0.0433
## outcome_grouped10-40:event_typeelection  0.7513     0.6063   1.239  0.2152
## outcome_grouped60-90:event_typeelection  1.0460     0.5272   1.984  0.0472
##
## (Intercept)          ***
## outcome_grouped10-40
## outcome_grouped60-90          ***
## event_typeelection      *
## outcome_grouped10-40:event_typeelection
## outcome_grouped60-90:event_typeelection *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) ot_10-40 ot_60-90 evnt_t o_10-40:
```

```
## otc_g10-40 -0.770
## otc_g60-90 -0.911  0.644
## evnt_typlct -0.535  0.411    0.432
## otc_10-40:_  0.454 -0.558   -0.366   -0.850
## otc_60-90:_  0.522 -0.414   -0.445   -0.977   0.830
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```